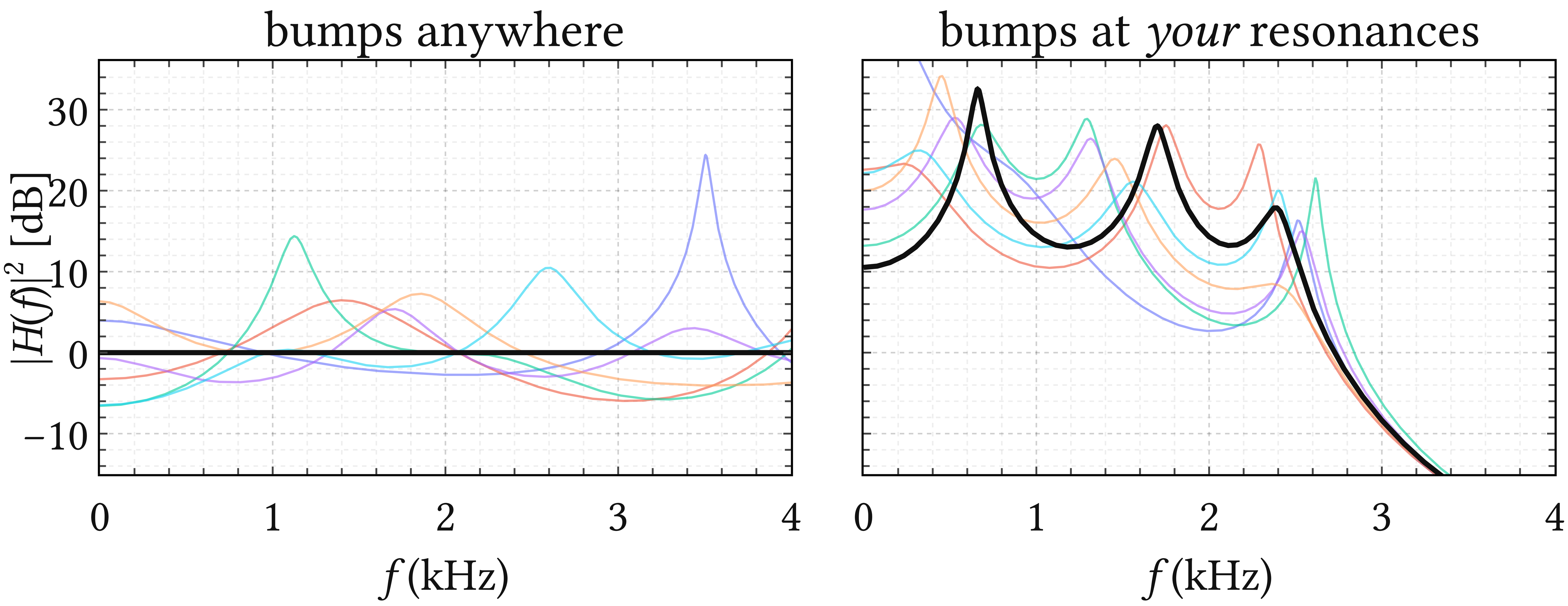
WANTED

ILL-POSED AUTOREGRESSIVE PROBLEMS



Envelope draws from a Gaussian AR(6) prior, before and after one closed-form D_{KL} update toward three formants (dotted); bold: prior mean.

THE MODEL

AR(P) with a Gaussian prior on the coefficients is conjugate: the posterior stays closed form.

$$x_t = a_1 x_{t-1} + \ldots + a_P x_{t-P} + \varepsilon_t$$

$$\mathbf{a} \sim \mathtt{Normal}(\boldsymbol{\mu}, \mathbf{S})$$

THE DICTIONARY

$A(z) = 1 - a_1 z^{-1} - \ldots - a_P z^{-P}$ shapes the envelope $|H|^2 = 1/\ |A|^2$; every expected feature is a factor:

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho\cos(\omega_0) z^{-1} + \rho^2 z^{-2}$	2
Real pole at ρ	$1 - \rho z^{-1}$	1
Tilt $-6k$ dB/oct	$(1 - \rho z^{-1})^k$	k
Seasonal pole of period s	$1 - z^{-s}$	s
k-fold resonance	$(1 - 2\rho\cos(\omega_0) z^{-1} + \rho^2 z^{-2})^k$	$2k$

WORKED EXAMPLE

AR(4), impose one resonance $Q(z) \mid A(z)$:

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$

$$Q(z) = 1 - c z^{-1} + d z^{-2}, \quad c = 2\rho\cos\omega_0, d = \rho^2$$

Divisibility is <u>linear</u> in $\mathbf{a}$:

$$Q \mid A \quad \Longleftrightarrow \quad \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0}$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it *on average*, $\mathbb{E}_{\mathbf{a}}[\mathbf{F}\mathbf{a} + \mathbf{c}] = \mathbf{0}$, by the minimum-$D_{\mathrm{KL}}$ update of $\mathtt{N}(\mathbf{0}, \lambda\mathbf{I})$: a Gaussian prior with the belief in its mean, uncertainty intact.

REWARD a prior with your knowledge in it, and a co-author who checks the math.

VRIJE UNIVERSITEIT BRUSSEL InViLab

Marnix Van Soom
AI Lab, Vrije Universiteit Brussel
marnix@ai.vub.ac.be

interactive figures